CHAPTER 07

Adapt when conditions change

A journey can become more demanding without any single dramatic event. Darkness, reduced grip, distraction and tiredness can combine. Reassess while you still have comfortable choices.

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Answers and review guidance

Official references

This material is a learning aid to help you understand Swedish driving theory. It is not an official source and cannot replace official information from Trafikverket and Transportstyrelsen. Always consult their current guidance and the applicable rules.

M07-L01

Drive within what you can see

Your goal: Choose a visibility and lighting strategy for darkness, fog and glare.

Requirements. Your speed must allow a stop within the visible road and before foreseeable hazards. Darkness does not create a separate allowance to travel beyond the distance you can assess. Keep the required lamps on; daytime running lights alone may be insufficient when conditions deteriorate. Traffic Regulation, chapter 3 §§14–15 and 68–71

Another changing condition: crosswinds

Independent crosswind warnings. The windsock shows the wind direction; neither example is a fog warning or a headlight instruction.



A24 · Crosswinds

Varning för sidvind

Strong crosswinds commonly occur on this stretch. The symbol is adapted to local conditions. Windsock extends towards the left.

Transportstyrelsen · official description



A24-2 · Crosswinds

Varning för sidvind

Strong crosswinds commonly occur on this stretch. The symbol is adapted to local conditions. Windsock extends towards the right.

Transportstyrelsen · official description

Continue the learning

Use main beam when needed for a sufficient field of view, but not where the road is adequately lit, where it can dazzle an approaching driver or an adjacent rail/ship operator, or at close distance behind another vehicle. Dipped beam is the usual driving light. In fog or heavy precipitation, front fog lights may replace dipped beam; the prohibited combinations must not be used together. A rear fog light is restricted to fog, heavy precipitation or comparable reduced visibility and must not dazzle a following driver. Traffic Regulation, chapter 3 §§69–72

Learning approach. Make lighting and speed one decision. If you dip the lights and can see less far, reassess speed immediately. Check the edges of the visible road for people and animals rather than assuming a dark area is empty. Keep windows and mirrors clear enough for a useful view and learn demisting controls before a difficult journey.

Automatic lighting is an aid, not confirmation that every required lamp is on. In a car whose daytime-light mode leaves the rear lights off, check the actual setting when darkness or bad weather arrives. Transportstyrelsen: vehicle lighting

Light and visual information

Lighting choice · Name the prohibited pair. Do not use dipped headlights and front fog lights together. In fog or heavy precipitation, front fog lights may replace dipped headlights; they are not an extra light to add to them. Select a permitted mode and check it on the actual vehicle. Transportstyrelsen: dipped beam and front fog lights

Central vision · Read the detail. Central vision supplies the fine detail where you are looking directly. Reading the precise wording on a road sign relies particularly on it. Merely noticing a sign at the edge of your view does not establish what its text says. Official driver-vision guidance

Peripheral vision · Notice, then inspect. Peripheral or side vision surrounds your direct focus. It can alert you to movement, a developing hazard or a traffic change outside that focus. Move your eyes, and head where needed, to inspect the relevant object directly while maintaining control and awareness of the road. For example, notice movement near the verge, then look to identify whether it is a person about to enter your path. Keep scanning; staring at one sign can leave other information unattended. Official driver-vision guidance

One light · Keep width uncertain. One approaching light at night does not prove a narrow motorcycle. A wider car can have a failed headlight. Transportstyrelsen documents defective car headlights; the practical inference is to allow for a vehicle whose full width is not yet visible. Preserve a safe lateral margin and adapt speed to the view rather than squeezing past on a guessed width. Transportstyrelsen: defective car headlights

Reassess the visible road



Rain and darkness shorten the usable view. The teal car travels on the right. Reassess speed and check the actual exterior-light setting; the illustration does not identify a lamp mode or a measured stopping distance.

Worked decision. Rain becomes heavy during a daytime trip. The dashboard remains illuminated, but the driver checks and selects the appropriate exterior lighting rather than inferring that the rear lights must be on. The driver also reduces speed for the shorter usable view.

Check your understanding

Knowledge check. (1) Can you maintain speed solely because the limit is unchanged after dipping main beam? (2) May dipped headlights and front fog lights be used together? (3) Which vision role reads precise sign text, and which can alert you to movement outside direct focus? (4) Does one approaching light prove a narrow motorcycle?

Remember: Adjust speed to the information the light actually gives you.

M07-L02

Grip is a changing resource

Your goal: Explain how surface conditions, tyres and driver inputs affect stability, steering and braking.

Tyres need friction to change the vehicle's speed and direction. Braking while cornering shares that capacity; ABS can help preserve steering forces during braking, but the available friction still limits the result. Water, snow and ice change the conditions at the tyre contact, so dry-road expectations cannot simply be carried into them. VTI friction report, sections 4.2–4.3

Warnings about grip

Independent slippery-road and loose-chippings warnings. Observe the actual surface; these warnings are not limited to winter.



A10 · Slippery road

Varning för slirig väg

The road may be slippery. Do not assume that the warning refers only to snow or ice. Skidding car inside a red-bordered yellow triangle.

Transportstyrelsen · official description



A11 · Loose chippings

Varning för stenskott

There is a risk of stones being thrown up. This warning is not normally used where that risk is routine, such as a gravel road. Car wheel throws stones inside a warning triangle.

Transportstyrelsen · official description

Continue the learning

Requirements for ordinary light cars and their trailers, checked 1 October 2026. Between 1 December and 31 March, winter tyres or equivalent equipment are required when winter conditions exist. Snow, ice, slush or frost on any part of the road counts; police determine conditions at the location. The minimum tread depth for the ordinary light-vehicle case is 3 mm. Studless winter tyres for the car need the mountain/snowflake marking. Winter tyres

Studded tyres are permitted from 1 October to 15 April and outside that period when winter conditions exist or are expected. Do not mix studded and non-studded tyres on an ordinary passenger car or light goods vehicle. In winter conditions, a studded towing car requires a studded trailer; the reverse combination is allowed. A light-trailer transition permits specified winter-designed M+S tyres through 30 November 2028. Exemptions and heavier-vehicle cases need their own checks; they are not the ordinary car rule. Winter tyres · Traffic Regulation, chapter 4 §§18a–18d

Check local studded-tyre restrictions

Seasonal permission to use studs does not override a local prohibition. This road sign is not a tyre sidewall marking.



C44 · No studded-tyre motor vehicles except class II mopeds

Förbud mot trafik med annat motordrivet fordon med dubbdäck än moped klass II

The prohibition concerns motor vehicles with studded tyres. Class II mopeds are included only if a supplementary panel says so. Studded tyre crossed by a red diagonal.

Transportstyrelsen · official description

For summer conditions, the minimum tread depth is 1.6 mm. Transportstyrelsen recommends placing the tyres with the best grip or greatest tread depth at the rear and checking pressure against the vehicle's instructions. A legal minimum is not a promise of equal performance in every condition. Transportstyrelsen: tyres

Read changing surfaces

Surface change · A bridge can freeze first. A bridge is exposed to cold air above and below. That exposure can cool it faster than neighbouring road supported by ground, so moisture can freeze on the bridge while the approach still appears clear. It is a possibility, not a rule that every bridge always freezes first. Anticipate uncertain grip and reduce speed before reaching the bridge. National Weather Service: bridge cooling

Rain on a cold road · Ice can hide. Rainwater can freeze when it meets a road surface cold enough for ice to form, producing a thin, nearly invisible slippery layer. The road may look wet, and the air temperature can be above zero while the surface is still below freezing. Near-zero air alone does not prove ice, but it calls for attention to surface conditions, more space and lower speed. Trafikverket: deceptive ice and cold roads

First rain after dry weather · Residues matter. During a dry spell, dust, dirt and oily residues can accumulate. The first rain can mix with these contaminants and make the road unusually slippery. Reassess speed and separation as rain starts; do not assume a light shower immediately washes the surface clean or that a fixed time restores grip. County Fire: rain after a dry spell

Worked surface decision. A light shower begins after dry weather as you approach a bridge in near-freezing conditions. Plan for both wet residues and possible local ice. The clear-looking approach does not measure the grip on the bridge.

Understand grip and handling

Rear stability · Why the best grip belongs at the rear. Transportstyrelsen recommends the best-grip or deepest-tread tyres at the rear to reduce skid risk in braking and bends, for both front- and rear-wheel-drive cars. Better rear grip helps reduce the risk of the rear sliding outward and rotating the car. Suitable tyres and stability assistance still cannot guarantee control. Transportstyrelsen: best tyres at the rear · VTI: rear grip and oversteer

Oversteer · Rear grip loss. In the loss-of-grip situation, the rear tyres lose sufficient lateral grip, the tail swings outward and the car rotates more than intended into the turn. This can develop into a spin. VTI: oversteer and rear stability

Understeer · Front grip loss. In the loss-of-grip situation, the front tyres cannot supply the lateral grip needed for the requested turn. The car turns less than intended and follows a straighter, wider path; adding more steering does not guarantee the requested curve. These are grip-loss explanations, not a full definition of every vehicle's normal handling characteristics. VTI: handling paths · NHTSA: front-axle grip limit

Aquaplaning · Water interrupts contact. If water builds between tyres and road, it can support the tyre on a film and reduce or remove effective tyre-road contact. Steering and braking then have less grip to work with. Reduce speed before standing water and uncertain wet surfaces. PennDOT: aquaplaning

Response · Ease off smoothly. If aquaplaning starts and there is space to slow gradually, ease off the accelerator, keep steering steady and avoid abrupt steering or hard braking. Make only gentle necessary corrections as contact returns. This is different from M02-L01's ABS emergency-stop instruction: ABS cannot create grip where a tyre is separated from the road by water. An immediate collision emergency needs action appropriate to the actual situation; this advice does not establish a universal ban on braking. PennDOT: aquaplaning response

Worked grip comparison. A rear slide swinging the tail outward is oversteer; a front slide that prevents the intended curve is understeer. A light, unresponsive steering feel over standing water raises a different possibility: aquaplaning. Recognise the mechanism, preserve margins and practise vehicle-specific emergency handling only with suitable instruction.

Prepare before grip changes

Learning approach. Reduce speed and increase space before reaching uncertain grip. Avoid demanding abrupt braking, steering and acceleration together. Do not experiment with skid recovery in ordinary traffic. Establish the actual vehicle's emergency-braking behaviour through appropriate instruction and risk training. Trafikverket: winter driving advice · Risk training



A dry-looking foreground leads towards a shaded winter bend. The surface appearance is a reason to reassess, not a measurement of friction. No skid-recovery technique or guaranteed tyre performance is shown.

Worked decision. Your car has suitable winter tyres, but the next bend is partly shaded and looks different from the road behind. You reduce speed before the bend and preserve margin. Equipment does not establish the grip at that particular point.

Check your understanding

Knowledge check. (1) Does legal tread depth guarantee dry-road spacing on ice? (2) Why can a bridge freeze first? (3) What can rain form on a sufficiently cold road? (4) Why can first rain after dry weather be unusually slippery? (5) Why place the best-grip tyres at the rear? (6) Compare oversteer and understeer in a grip-loss situation. (7) What is aquaplaning, and how should you respond when there is room to slow gradually?

Remember: Tyres and assistance systems support a cautious plan; they do not create unlimited grip.

M07-L03

Protect your attention

Your goal: Recognise distraction, stress and fatigue and choose an interruption or alternative travel plan.

Requirements. During road travel, a driver may use communication equipment only if it does not adversely affect driving. Holding the equipment in the hand while using it is prohibited. Hands-free use is therefore not automatically acceptable: the effect on driving still matters. You must also not drive when exhaustion or another condition prevents safe control. Transportstyrelsen: communication equipment · Traffic Regulation, chapter 3 §1 and chapter 4 §10e

Learning approach. Prepare navigation before setting off, place the phone away from casual handling and agree when passengers can help. If a conversation, screen task or emotional reaction is consuming the attention needed for the road, end it. A safe place to stop is the place to resolve a complicated route or urgent message.

Attention link. Use M07-L01's notice-then-inspect distinction when scanning. A conversation that consumes attention can delay recognising what peripheral vision has alerted you to.

Treat fatigue as a reason to change the journey. Plan rest opportunities and stop safely when concentration cannot be maintained. Do not wait for a dramatic event to prove that you are too tired. Trafikverket recommends setting out rested, allowing time and taking breaks on longer journeys. No universal break interval in this material certifies that you can continue safely. Journey advice

Plan a safe rest stop

A rest-area destination can support journey planning. The sign gives no prescribed break interval or assurance that a tired driver can safely continue.



H13 · Rest area

Rastplats

This identifies a rest area with a toilet accessible to people with mobility impairments. Picnic table and tree.

Transportstyrelsen · official description

Protect attention before moving



A left-hand-seat driver is parked in a parking area, with the phone in a holder. This is a stopped discussion, not permission to use a screen or conduct a distracting call while driving. Controls and mounting are illustrative.

Worked decision. A driver misses a sign during an intense hands-free conversation. The useful response is not to argue that the device is legal to use. End the conversation, reassess fitness and stop safely if attention cannot be restored.

Check your understanding

Knowledge check. Is entering a destination on a mounted phone necessarily allowed because it is not hand-held?

Remember: Protect attention before it is fully occupied.

M07-L04

Change the plan when risk grows

Your goal: Reassess a trip when weather, darkness, delays or passenger pressure accumulate.

Learning approach. At a meaningful change—worsening visibility, an unexpected closure, a missed rest opportunity—review the whole plan. Ask what has changed in the driver, vehicle, route and time available. A manageable single difficulty may become demanding when several occur together. This is a planning exercise, not a formula that assigns a numerical risk score.

Choose between continuing with a suitable margin, stopping safely, changing route, changing driver if lawful and suitable, or cancelling the journey. The choice depends on the actual conditions. A driver unable to maintain safe control must not continue merely because the destination is close. Traffic Regulation, chapter 3 §§1 and 14

The syllabus treats route choice, timing, passengers, fatigue and personal motives as parts of driving competence. Use the same questions after a journey: Which warning did I notice early? Which did I explain away because I wanted to finish? Syllabus, chapters 4–5

Reconsider the whole journey



Two adults reconsider a journey indoors while rain falls outside and the car remains parked. Use the available route information to choose a realistic alternative before setting off.

Worked decision. A late appointment leaves a driver facing an unfamiliar route in darkness and worsening weather. Instead of protecting the original arrival time, the driver finds a safe stopping place, tells the person waiting and chooses a realistic alternative. The change happens while there is still time to make it calmly.

Check your understanding

Knowledge check. Is deciding to cancel evidence that the original plan failed?

Remember: Replanning is a driving skill.

Review and self-assessment

Scenario: You leave in daylight but meet heavy rain, darkness and a delay. A passenger wants to keep going without a break.



Departure: identify what to prepare before leaving in daylight.



Rain: explain which observations would change your decisions.

Describe the changes you notice and the information you would check.

Darkness, delay and pressure



Darkness: explain how the changed view affects the journey.



Delay: consider the pressure to keep going. The clock is illustrative, not a prescribed break interval.

Explain changes to lighting, speed, space and the journey itself.

Keep a practice record

Record the earliest sign that would make you change this trip. Name the consequence of ignoring it and one action you can prepare in advance. Syllabus self-assessment, chapters 2–4 §3

Review after answering

M07-L01

Answer: (1) No; reassess safe speed for the view. (2) No. Front fog lights may replace dipped beam in fog/heavy precipitation. (3) Central vision reads fine detail; peripheral vision can detect movement/hazards. Look directly to inspect. (4) No. It may be a wider car with a failed lamp; preserve clearance.

M07-L02

Answer: (1) No; adapt speed and gap. (2) Cold air cools it above and below. (3) A nearly invisible slippery ice layer. (4) Water mixes with accumulated dirt/oily residues. (5) To reduce rear grip loss, tail swing and rotation. (6) Oversteer: rear grip loss, tail outward, more rotation than intended. Understeer: front grip loss, less turn and a wider/straighter path. (7) A water film reduces effective tyre-road contact. Ease off the accelerator, keep steering steady and avoid abrupt steering/hard braking; use gentle necessary corrections.

M07-L03

Answer: No. Equipment use must not adversely affect driving. Prepare it while parked or stop somewhere safe and lawful.

M07-L04

Answer: It may show that reassessment worked. The aim is a safe decision under current conditions, not loyalty to an earlier plan.

M07-R

Review guide: Check what you can see, what grip you can reasonably expect and whether you can concentrate. Do not compensate for delay by reducing margins.

Official references

Original references checked 5 October 2026; G7 additions checked 13 October 2026. Sign captions and artwork use the verified official asset register. Consult current guidance and applicable rules.

Traffic Regulation, chapter 3 §§14–15 and 68–71

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Transportstyrelsen: vehicle lighting

https://www.transportstyrelsen.se/sv/vagtrafik/trafikregler-och-vagmarken/trafikregler/generella-trafikregler/fordonsbelysning/

Transportstyrelsen: dipped beam and front fog lights

https://www.transportstyrelsen.se/globalassets/global/publikationer-och-rapporter/vag/trafikant/produkter/tran-049-choose-the-right-vehicle-lighting_a5_webb.pdf

Official driver-vision guidance

https://www.dmv.ca.gov/portal/driver-education-and-safety/medical-conditions-and-driving/vision-conditions/

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Winter tyres

https://www.transportstyrelsen.se/vinterdack

Transportstyrelsen: tyres

https://www.transportstyrelsen.se/sv/vagtrafik/fordon/fordonsregler/dack/

Trafikverket: winter driving advice

https://www.trafikverket.se/resa-och-trafik/trafiksakerhet/sakerhet-pa-vag/dack/

Risk training

https://www.transportstyrelsen.se/sv/vagtrafik/korkort/ta-korkort/riskutbildning/riskutbildning-bil/

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National Weather Service: bridge cooling

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Trafikverket: deceptive ice and cold roads

https://www.trafikverket.se/om-oss/nyheter/lansnyheter/stockholm/2025/2025-11/pa-glid-nej-sa-jobbar-vi-mot-halka-i-stockholmstrakten/

County Fire: rain after a dry spell

https://main.sbcounty.gov/2019/12/04/arrive-safely-with-these-winter-driving-safety-tips/

VTI: rear grip and oversteer

https://fudinfo.trafikverket.se/fudinfoexternwebb/Publikationer/Publikationer_001601_001700/Publikation_001666/overstyrn-pa-is-o%20sno-med-olika-vinterdack.pdf

NHTSA: front-axle grip limit

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PennDOT: aquaplaning

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Transportstyrelsen: communication equipment

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Journey advice

https://www.trafikverket.se/resa-och-trafik/trafiksakerhet/sakerhet-pa-vag/goda-rad-for-bilkorning-vid-storhelger/

Syllabus, chapters 4–5

https://www.transportstyrelsen.se/TSFS/TSFS%202011_20.pdf

Road-sign illustrations: Transportstyrelsen. Each reference example links to its official description. Swedish names are official; English labels and explanations are editorial translations. Road scenes are simplified illustrations, not to scale.